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## Unit “ 2 : Session “ 10 : SLO “ 2 | Doubly Linked List “ Insertion at the Middle

### ■ I. Insert a New Node VAL After a Given Node NUM

- Step 1: IF AVAIL = NULL
- Write OVERFLOW
- Go to Step 12
- [END OF IF]
- Step 2: SET NEW\_NODE = AVAIL
- Step 3: SET AVAIL = AVAIL -> NEXT
- Step 4: SET NEW\_NODE -> DATA = VAL
- Step 5: SET PTR = START
- Step 6: Repeat Step 7 while PTR -> DATA != NUM
- Step 7: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 8: SET NEW\_NODE -> NEXT = PTR -> NEXT
- Step 9: SET NEW\_NODE -> PREV = PTR
- Step 10: SET PTR -> NEXT -> PREV = NEW\_NODE
- Step 11: SET PTR -> NEXT = NEW\_NODE
- Step 12: EXIT

### ■ II. Insert a New Node VAL Before a Given Node NUM

- Step 1: IF AVAIL = NULL
- Write OVERFLOW
- Go to Step 12
- [END OF IF]
- Step 2: SET NEW\_NODE = AVAIL
- Step 3: SET AVAIL = AVAIL -> NEXT
- Step 4: SET NEW\_NODE -> DATA = VAL
- Step 5: SET PTR = START
- Step 6: Repeat Step 7 while PTR -> DATA != NUM
- Step 7: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 8: SET NEW\_NODE -> NEXT = PTR
- Step 9: SET NEW\_NODE -> PREV = PTR -> PREV
- Step 10: SET PTR -> PREV -> NEXT = NEW\_NODE
- Step 11: SET PTR -> PREV = NEW\_NODE
- Step 12: EXIT